



POLITECNICO
MILANO 1863

High Performance Scientific Computing for Aerospace Project

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Structure of the input file:

- **Mesh points:** Number of points in X, Y, Z directions
 - ▶ Example: 50 50 50 $\Rightarrow N_x = 50, N_y = 50, N_z = 50$
- **Time parameters:**
 - ▶ dt - Time step size, e.g., 0.01
 - ▶ T - Total simulation time, e.g., 10.0
- **Spatial step sizes:**
 - ▶ dx dy dz - Grid spacing in X, Y, Z, e.g., 0.5 0.5 0.5
- **Initial conditions files:**
 - ▶ u0_init_file - File with initial velocity values
 - ▶ p0_init_file - File with initial pressure values
 - ▶ k_file - File with permeability values

- Computation of grid.
- Input values κ , ν , initial conditions handling.
- Establish the order of nodes to maximize performance.
- Potentially excessive memory usage.

- Optimization of the storage in data structures.
- Complete the definition of the data structures for β and γ .
- Decide how to deal with input data κ , ν , initial conditions (if they are a given function or a set of data).
- Discuss about the parallelization strategy for the solution of linear systems.
- Actual implementation of code.